

§ 10.4 Calculus in Polar Coordinates (극좌표에서의 미적분)

Ex.2 까지

■ Area

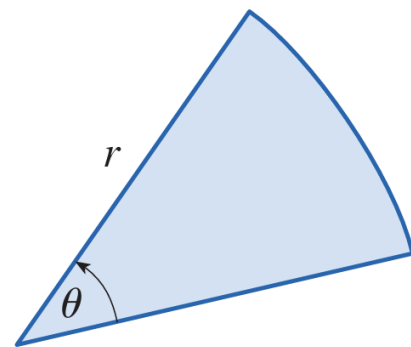
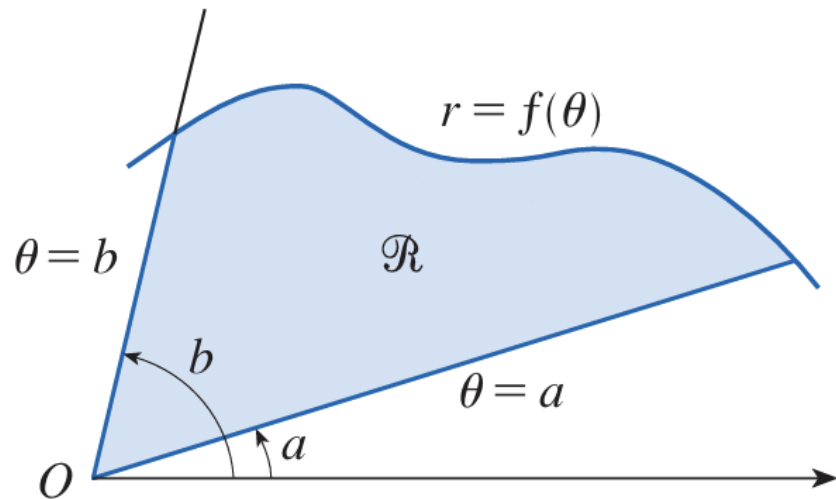


FIGURE 1

Let $\mathcal{R}$ be the region, illustrated in Figure 2, bounded by the polar curve $r = f(\theta)$ and by the rays $\theta = a$ and $\theta = b$, where f is a positive continuous function and where $0 < b - a \leq 2\pi$.

**FIGURE 2**

the formula for the area A of the polar region $\mathcal{R}$

3

$$A = \int_a^b \frac{1}{2} [f(\theta)]^2 d\theta$$

Formula 3 is often written as

4

$$A = \int_a^b \frac{1}{2} r^2 d\theta$$

EXAMPLE 1 Find the area enclosed by one loop of the four-leaved rose $r = \cos 2\theta$.

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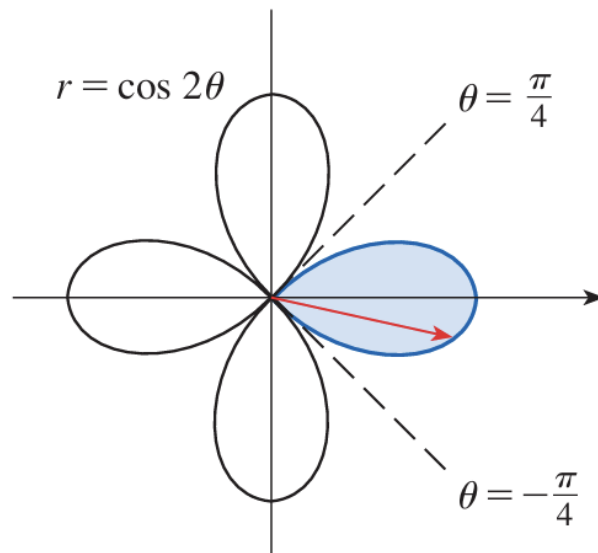


FIGURE 4

EXAMPLE 2 Find the area of the region that lies inside the circle $r = 3 \sin \theta$ and outside the cardioid $r = 1 + \sin \theta$.

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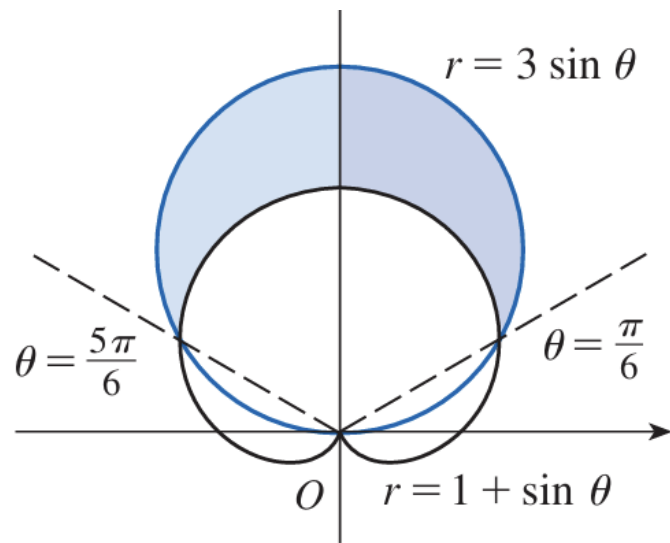


FIGURE 5

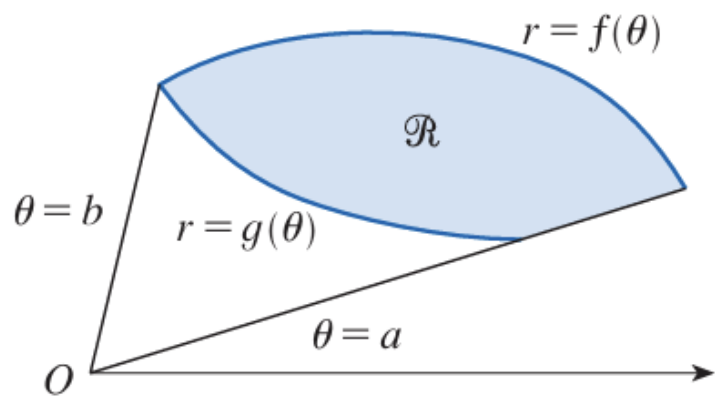


FIGURE 6

$$\begin{aligned}
 A &= \int_a^b \frac{1}{2} [f(\theta)]^2 d\theta - \int_a^b \frac{1}{2} [g(\theta)]^2 d\theta \\
 &= \frac{1}{2} \int_a^b ([f(\theta)]^2 - [g(\theta)]^2) d\theta
 \end{aligned}$$

